AITOGY-GNSS-BASE-ESP-UBX

H ng d n kh ch h ng | Firmware V2.3.5 | C p nh t 25/10/2026

1. Ph m vi v nguy n t c v n h nh

B AITOGY-BASE-ESP nh n d li u t receiver GNSS, chuy n ti p RTCM l n caster b ng 4G ho c Wi-Fi v nh n i u khi n t h th ng CORS. Anten base ph i ng y n khi ph t c i ch nh. Sau khi di chuy n anten, c u h nh l i t a base tr c khi cho rover s d ng.

T i li u m t firmware **V2.3.5**, c p nh t **25/10/2026**. K t qu bi n d ch v ki m tra ph n m m kh ng thay th nghi m thu ngo i th€c a. Kh ng cam k t th•i gian kh , i ng c nh hay ch nh x c o khi ch a ki m tra receiver, anten, m ng c i ch nh v h t a .

Ch	Khi s d ng	i u ki n ho n th nh
FIXED_LLA	ƒ„ c… t a i m t anten ng tin c y.	G i c u h nh fixed, ki m tra t a receiver v lu†ng RTCM.
SURVEY_IN	C n receiver t€ kh o s t v tr ng y n.	Receiver k t th‡c kh o s t; g i l nh th nh c ng kh ng c… ngh‚a kh o s t „ xong.
AUTO_CORS	C… t i kho n rover nh n c i ch nh t CORS.	Chuy n rover, nh n c i ch nh, l y m‰u RTK_FIXED, t l i base v ki m tra ph t RTCM.

ƒ cao ph i dŠng ‡ng quy c c‹a t a cung c p. V i c u h nh fixed theo LLA, s d ng cao ellipsoid. Kh ng thay tr€c ti p b ng cao th‹y chu n ho c b• qua chi u cao anten/ i m quy chi u.

2. K t n i v chu n b

ng t n hi u TDM240x	K t n i
GPIO18 / RX GNSS c‹a ESP32	Nh n t TX c‹a receiver GNSS.
GPIO19 / TX GNSS c‹a ESP32	G i v o RX c‹a receiver GNSS.
GND	Chung mass gi a ESP32, receiver v b ngu†n theo thi t k b thi t b .
GPIO16 / RX modem	Nh n t TX modem.
GPIO17 / TX modem	G i v o RX modem.
Serial console	115200 baud; g i m t dŽng l nh v k t th‡c b ng LF.
UART GNSS	M c nh 115200 baud, 8N1.

ƒ i chi u nh„n c•ng v s• † c‹a ‡ng board tr c khi u d‘y. Kh ng suy ra ch‘n c’m hay m c ngu†n t h nh nh c‹a board kh c. G’n anten GNSS ‡ng c•ng, t n•i tho ng tr•i, c nh c• kh ; ki m tra SIM/APN v anten 4G tr c khi ch y auto base.

B n u-blox dŠng receiver h“ tr” c c l nh c u h nh m firmware t o, m c ti u l dŽng ZED-F9P phŠ h”p. N i UART1 receiver t i RX18/TX19 ESP32. M t s receiver/phi n b n ch• h“ tr” m t ph n CFG-VALSET ho c TMODE3; kh ng suy lu n m i receiver u-blox u t •ng th ch.

3. C u h nh m c nh v th ng tin c n b n giao

Th ng s	Gi tr / ngh a trong firmware
K t n i m ng	4G; c… th chuy n Wi-Fi b ng ATG.
APN	v-internet; thay theo SIM th€c t .
MQTT broker m c nh	45.117.179.134:1883; ph i kh p h th ng CORS ang tri n khai.
Backend WebSocket	base.aitogy.com:8000, giao th c ws; c n endpoint thi t b th€c s€ truy c p ”c.
Caster d€ phŽng trong c u h nh m c nh	aitogy.com.vn:2101, mountpoint /test; kh ng ph i th ng tin c p ph t cho m i kh ch h ng.
K nh qu n l– ch nh	Topic theo serial esp_... do thi t b t o; qu n tr vi n c p quy n tr n CORS.
Tr ng th i backend	Chu k– m c nh 5 gi‘y; k t n i m ng c… th l m ch m vi c g i.
Health ki u c˜	Chu k– m c nh 30 gi‘y.

f •n v tri n khai b n giao: serial, t n tr m, lo i GNSS, caster/mountpoint ph t, t i kho n c i ch nh u v o n u dŠng AUTO_CORS, APN ho c Wi-Fi v ng •i qu n l– tr n CORS. M t kh Œu/token do qu n tr vi n c p ri ng, kh ng l y t v d trong t i li u.

4. a thi t b l n CORS

1. C p ngu†n •n nh, m , console 115200 baud v ghi l i serial esp_..., phi n b n firmware, lo i chip, baud nh n di n ”c.
2. C u h nh m ng, MQTT v backend theo tri n khai. C c chu“i trong d u <...> l gi tr c n thay, kh ng nh p nguy n d u ngo c.
3. L u th ng tin tr c, g i l nh ch n m ng sau cŠng v l nh … kh , i ng l i thi t b .
4. Tr n CORS, qu n tr vi n nh n/c p thi t b , t t n, tri n khai license v ch n ‡ng lo i chip.
5. Tri n khai d ch v caster: server 1, server 2 n u c n, mountpoint, t i kho n ph t v ch lu n ph t/theo y u c u.
6. Ch n FIXED_LLA, SURVEY_IN ho c AUTO_CORS. Theo d™i k t qu l nh v ti n , sau … ki m tra lu†ng RTCM b ng rover c l p.

V d c u h nh 4G:

```
ATG ESP SET 4G APN <apn>
ATG MQTT SET SERVER <mqtt_host>
ATG MQTT SET PORT 1883
ATG MQTT SET USER <mqtt_user>
ATG MQTT SET PASS <mqtt_password>
ATG ESP SET BACKEND HOST <backend_host>
ATG ESP SET BACKEND PORT 8000
ATG ESP SET CONNECTION 4G
```

V d •i sang Wi-Fi:

```
ATG ESP SET WIFI SSID <ssid>
ATG ESP SET WIFI PASS <wifi_password>
ATG ESP SET CONNECTION WIFI
```

Console ATG t ch tham s theo d u c ch, kh ng h“ tr” chu“i c… kho ng tr’ng ho c gi tr r“ng. N u SSID/m t khŒu c n d ng n y, li n h kš thu t; kh ng dŠng d u nh y gi l p chu“i.

Thi t b ch a ”c c p quy n ho c ch a c… license v‰n c… th b o tr ng th i nh ng kh ng ‹ i u ki n ph t RTCM. Tr ng th i online c‹a k nh qu n l– c˜ng ch a ch ng minh rover nh n ”c c i ch nh.

5. Ch n ch base v x c nh n k t qu

5.1 T a c nh

Nh p vˆ /kinh d ng th p ph‘n v cao ellipsoid t nh b ng m›t. Ki m tra ‡ng i m quy chi u anten v h t a . Sau khi c u h nh, i chi u t a c t receiver, s€ hi n di n c‹a b n tin t a base v t a rover t i i m ki m tra.

5.2 Survey-in

f t anten ng y n, nh p th•i gian kh o s t theo gi‘y. Nh nh u-blox c u h nh TMODE3 v i c th•i gian t i thi u v gi i h n ch nh x c. Receiver t€ quy t nh k t th‡c survey theo i u ki n c u h nh; theo d™i tr ng th i th€c t sau khi g i.

Survey-in kh ng t€ t o ra t a tuy t i t ch nh x c kh o s t ch• v ch n ng œng nh•. Khi c n t a g’n v i h quy chi u c th , dŠng i m „ bi t ho c AUTO_CORS v th€c hi n ki m tra c l p.

5.3 Auto base t€ CORS

Firmware chuy n receiver sang rover, nh n RTCM t mountpoint c i ch nh v ch• ch p nh n GGA m i v i tr ng th i **RTK_FIXED**. RTK_FLOAT kh ng ”c dŠng l m m‰u auto base.

Lu†ng m c nh c n 5 l n ki m tra FIX li n ti p theo nh p kho ng 1 gi‘y, sau … l y t i a 60 m‰u trong c a s• 60 gi‘y. C n t nh t 15 m‰u h”p l ti p t c; kh ng m c nh coi k t qu lu n c… ‹ 60 m‰u. Thi t b t l i t a base r†i kh i ph c d ch v ph t „ c u h nh.

Th•i gian ch• FIX m c nh 3600 gi‘y; qu n tr vi n c… th c u h nh theo nghi p v . Khi m t FIX, m t k t n i, thi u m‰u ho c c… l nh d ng, xem l– do trong ti n workflow. Vi c kh i ph c k t n i caster sau th t b i kh ng t€ ch ng minh receiver „ tr , v c u h nh base mong mu n.

Ph n chuy n VN2000 hi n dŠng tham s t ch h”p v ph›p chi u theo l€a ch n t•nh/kinh tuy n. Firmware kh ng gi i m„ c c b n tin datum RTCM 1021/1023/1025. Ph i i chi u m c ki m tra th€c a; kh ng coi nh„n VN2000 l ch ng nh n ‡ng h t a .

6. Theo d•i tr ng th i v lu , ng d ƒ li u

Hi„n th /tr ng	C ch hi„u
awaiting_license / unprovisioned	Ch a ‹ i u ki n c p quy n/license ph t RTCM.
locked	Thi t b b kh…a to n b ; li n h qu n tr vi n.
sleep	ƒ ang ch• k ch ho t lu†ng theo y u c u; ch a ch'c l m t m ng.
online	K nh qu n l– ho t ng; c n ki m tra th m t ng server NTRIP.
auto_setup_* / reference_check*	Workflow ang th€c hi n m t b c c u h nh ho c ki m tra.
last_command_result	K t qu x l– l nh; l nh kh , i ng workflow th nh c ng ch• ngh^a „ nh n vi c.
auto_base_progress	Theo d™i phase, s m‰u v l– do completed/failed/stopped c‹a workflow.
server1/2_connected v server1/2_bps	K t n i v s byte/gi'y g i t i t ng caster.
rtcm_input_bps	S byte RTCM/gi'y "c …ng khung , UART; kh ng ph i ch ng nh n CRC h"p l .
GGA/fix/satellites	Ch t l "ng nghi m GNSS v d li u v tr ang nh n.

V d c u tr‡c health ki u c˜ (gi tr minh h a):

```
{
  "uptime_s": 120,
  "free_heap_bytes": 120000,
  "connected_via": "GSM",
  "rssi_dbm": 20,
  "mqtt_ok": true,
  "ntrip_ok": true,
  "gnss_data_ok": true,
  "rtcm_decoder": false,
  "rtcm_types": {}
}
```

connected_via dŠng GSM ho c WiFi. gnss_data_ok ph n nh „ nh n khung RTCM g n 'y, trong ng œng 5 gi'y c‹a t c v health. rtcm_types r"ng v ESP32 kh ng m t ng lo i b n tin. rssi_dbm l t n tr •ng c˜: Wi-Fi dŠng RSSI, cŽn 4G ang gi gi tr ch t l "ng t n hi u th do modem tr v , ch a t€ chuy n sang dBm. Kh ng dŠng gi tr … nh s dBm 4G „ hi u chu Œn.

H th ng nh n NMEA GGA/GSA/GSV/GST theo d™i GNSS. ESP32 kh ng m c nh t o m t b n sao to n b RTCM l n MQTT. C c tr •ng c… t n valid_packets hi n l b m …ng khung, kh ng c… ki m tra CRC RTCM. hacc c… th l y t GST ho c fallback HDOP; kh ng coi m i gi tr hacc u l ch nh x c m›t „ x c nh n.

7. Danh m c l nh ATG

ATG dŠng c u h nh k t n i v h" tr" kš thu t. •u ti n CORS c u h nh base v c k t qu l nh theo t ng thi t b . T kh…a dŠng ‡ng ch hoa; m"i dŽng k t th‡c b ng LF. L nh SET th •ng l u c u h nh, c n kh , i ng l i k t n i c l i gi tr m i.

7.1 ESP v m ng

C… ph p	T c d ng
ATG ESP RESTART	Kh , i ng l i ngay.
ATG ESP SET CONNECTION 4G	L u ch n 4G v kh , i ng l i.
ATG ESP SET CONNECTION WIFI	L u ch n Wi-Fi v kh , i ng l i.
ATG ESP SET WIFI SSID <ssid>	L u SSID kh ng c… kho ng tr ' ng.
ATG ESP SET WIFI PASS <password>	L u m t khŒu kh ng c… kho ng tr ' ng.
ATG ESP SET 4G APN <apn>	L u APN.
ATG ESP SET 4G USER <user>	L u t n žng nh p APN khi nh m ng y u c u.
ATG ESP SET 4G PASS <password>	L u m t khŒu APN khi nh m ng y u c u.
ATG ESP SET BACKEND HOST <host>	L u hostname/IP c‹a endpoint WebSocket, kh ng kŸm scheme ho c •ng d‰n.
ATG ESP SET BACKEND PORT <port>	L u c•ng WebSocket; ch p nh n 1-65535.

7.2 MQTT v NTRIP ph t

C… ph p	T c d ng
ATG MQTT SET SERVER <host>	L u broker MQTT.
ATG MQTT SET PORT <port>	L u c•ng MQTT; nh p s h ” p l 1-65535.
ATG MQTT SET USER <user>	L u t n žng nh p broker.
ATG MQTT SET PASS <password>	L u m t khŒu broker.
ATG MQTT SET PUBTPCHEALTH <topic>	L u topic health ki u c˜.
ATG MQTT SET SUBTPCCMD <topic>	L u topic subscribe ATG ki u c˜; thay topic c n kš thu t ki m tra callback th€c nh n.
ATG NTRIP SET CSTRADDR <host>	L u caster m c nh/d€ phŽng.
ATG NTRIP SET CSTRPORT <port>	L u c•ng caster; nh p s h ” p l 1-65535.
ATG NTRIP SET MNTPNT <mountpoint>	L u mountpoint m c nh.
ATG NTRIP SET CSTRAUTH <password>	L u m t khŒu source nguy n vžn; kh ng nh p base64 c‹a username:password.

D ch v „ c p t CORS c… th u ti n h•n c u h nh NTRIP m c nh l u b ng ATG. Mu n thay t i kho n/mountpoint ang v n h nh, ch•nh d ch v tr n CORS r†i tri n khai l i.

7.3 GNSS v t l i c u h nh

C… ph p	T c d ng
ATG GNSS ROVER	Chuy n receiver sang rover ch Œn o n; kh ng t€ žng nh p t i kho n rover CORS.
ATG GNSS BASE SURVEY_IN <seconds> <accuracy_m>	G i batch survey-in theo lo i chip; m c nh ch• xu t RTCM, kh ng xu t NMEA diagnostic.
ATG GNSS BASE SURVEY_IN <seconds> <accuracy_m> W_DIAG	Survey-in v xu t c RTCM l‰n c c b n tin NMEA diagnostic; W_DIAG kh ng ph ‘n bi t hoa th •ng.
ATG GNSS BASE FIXED <lat> <lon> <alt> <accuracy_m>	G i t a LLA c nh; cao t nh b ng m›t ellipsoid.
ATG GNSS BASE <message> <port> ON	B t b n tin RTCM tr n c•ng ch• nh.
ATG GNSS BASE <message> <port> OFF	T ’t b n tin RTCM tr n c•ng ch• nh.
ATG GNSS SET CHIP UBLOX	L u l€a ch n chip v kh , i ng l i; boot v‰n th€c hi n nh n di n ph n c ng.
ATG CONFIG RESET	ƒ t c• n p l i m c nh v kh , i ng l i; c… th l m •i m ng v c u h nh GNSS.

C•ng l nh RTCM h“ tr ” I2C/UART1/UART2/USB/SPI; •ng n i ESP32 trong c u h nh chuŒn l UART1. V d : ATG GNSS BASE 1074 UART1 ON. B n tin ch a c… trong b ng l nh firmware c… th b t ch i dŠ receiver t€ n… h“ tr ”.

L nh ATG ki u c˜ ch‹ y u tr log Serial, kh ng c… k t qu JSON ri ng theo command_id nh l nh CORS. Kh ng coi dŽng „ g i¡ l receiver „ ch p nh n c u h nh. Reset c u h nh c c b kh ng thay th thao t c gœ thi t b /thu h†i quy n tr n CORS; qu n tr vi n ph i x l– tr ng th i c p ph t khi c n.

8. RTCM, ephemeris v c u h nh receiver

8.1 B n tin v gi i h n h† tr‡

B ng l nh RTCM c c b hi n c… 1005, 1074, 1084, 1094, 1124, 1077, 1087, 1097 v 1230. Nh nh u-blox ch a c… b t o l nh ephemeris 1019/1020/1042/1044/1045 nh Unicore; kh ng qu ng b t nh nžng … ch• v firmware dŠng chung ki u Command.

Survey/fixed dŠng UBX-CFG-TMODE3; output dŠng CFG-VALSET/CFG-MSG v l u CFG-CFG. Firmware nh tuy n c c output USB/UART2 trong batch backend v UART1 v t nh l i checksum. C n ki m tra receiver th€c s€ h“ tr ” c c message/key ”c c p.

8.2 Ki„m tra tr^c ti p b‰ng u-center

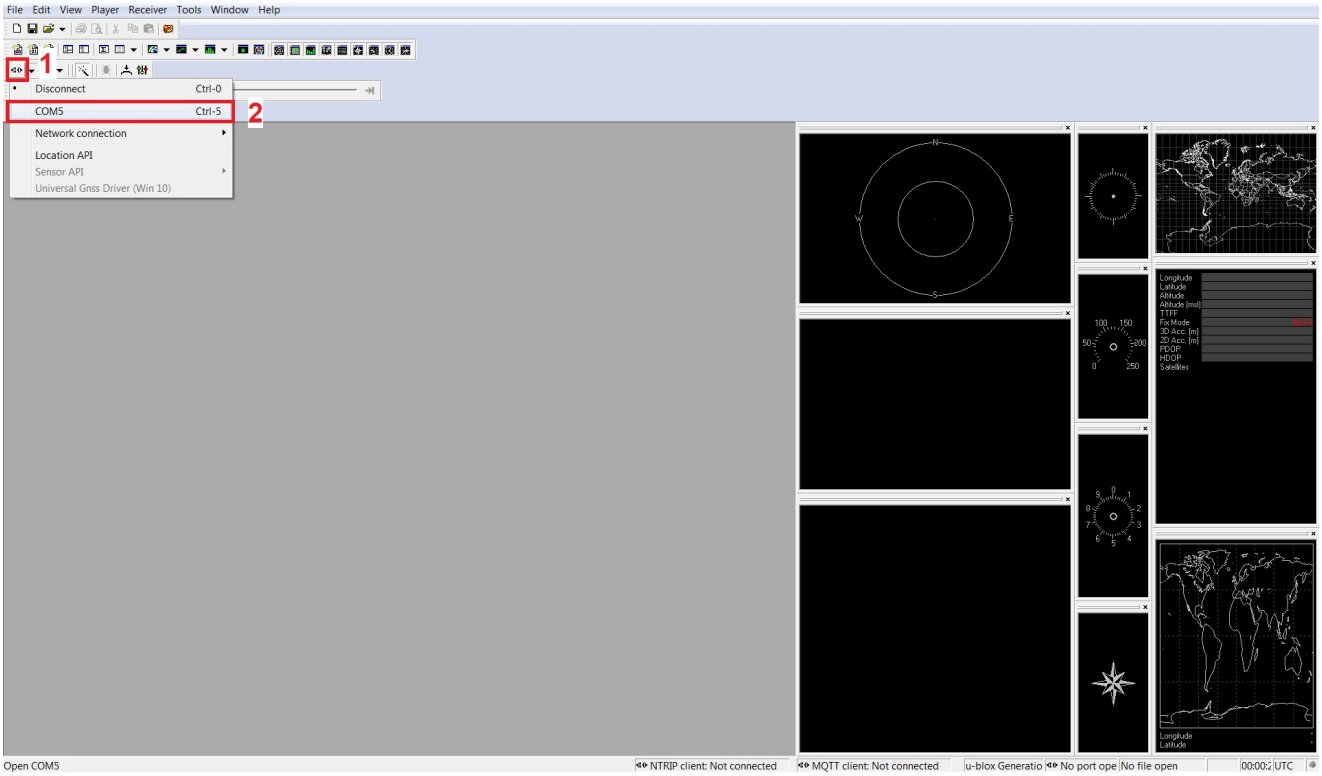
DŠng c ng c c u h nh u-blox phŠ h ”p model v phi n b n receiver. B nh d i ‘y minh h a giao di n u-center c• i n „ c… trong d€ n; t n menu c… th kh c , phi n b n kh c. K t n i ‡ng c•ng USB receiver, x c nh n model/version v baud tr c khi thay c u h nh.

1. Ch n k t n i receiver v dŠng t€ dŽ baud khi ch a bi t t c hi n t i.
2. M , Packet Console/Message View ph ‘n bi t NMEA, UBX v RTCM; RTCM kh ng ph i vžn b n thu n.
3. C u h nh UART1 l c•ng n i ESP32, ki m tra baud 115200 v protocol input/output phŠ h ”p.
4. Trong c u h nh Time Mode/TMODE3, ch n survey-in ho c fixed v nh p ‡ng •n v .

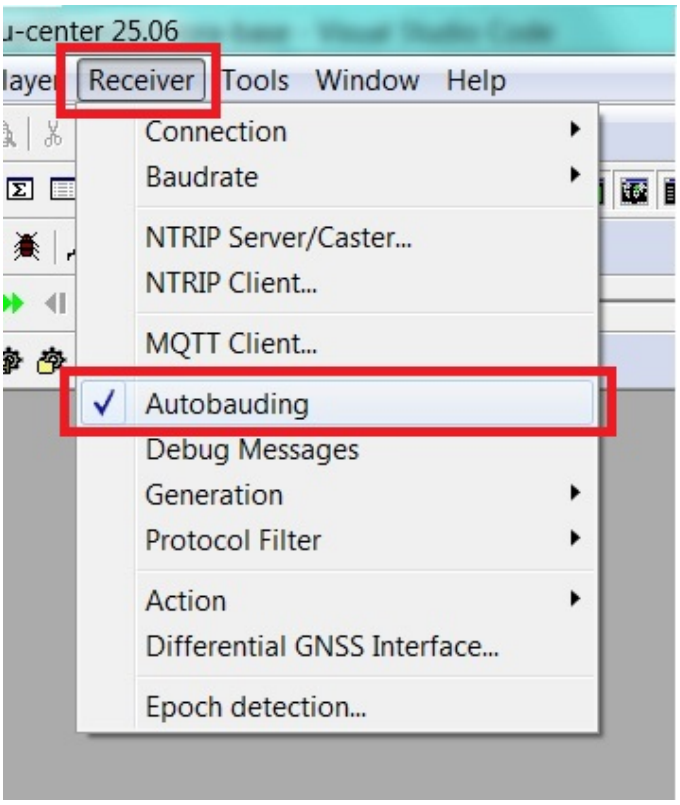
5. B t b n tin c n thi t , UART1, g i l nh, l u v o b nh receiver v ki m tra l i sau m t ngu†n.

Kh ng g i chu“i ASCII MODE BASE c‹a Unicore v o receiver u-blox. u-blox dŠng g…i UBX nh ph‘n v ki m tra ACK/NAK phŠ h”p.

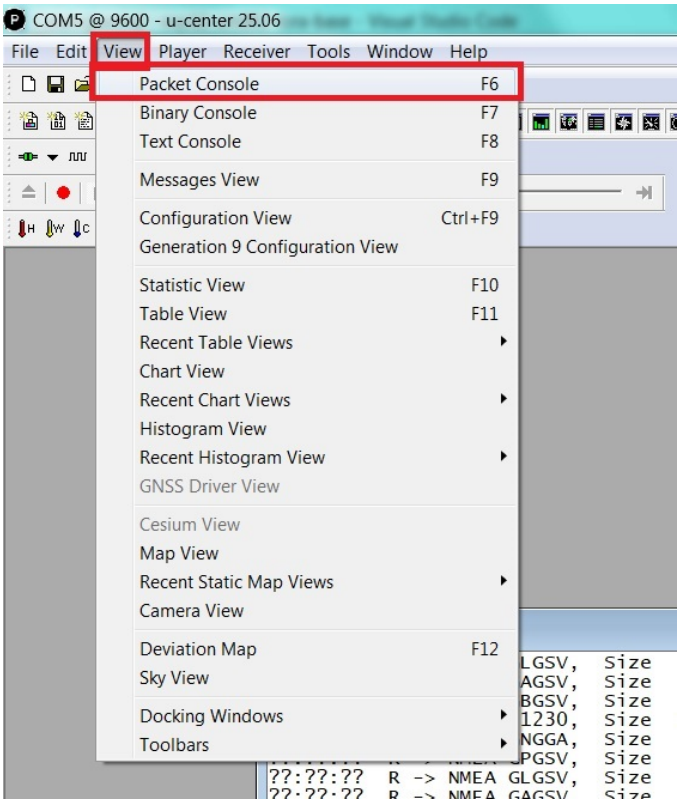
8.3 H nh minh h a u-center



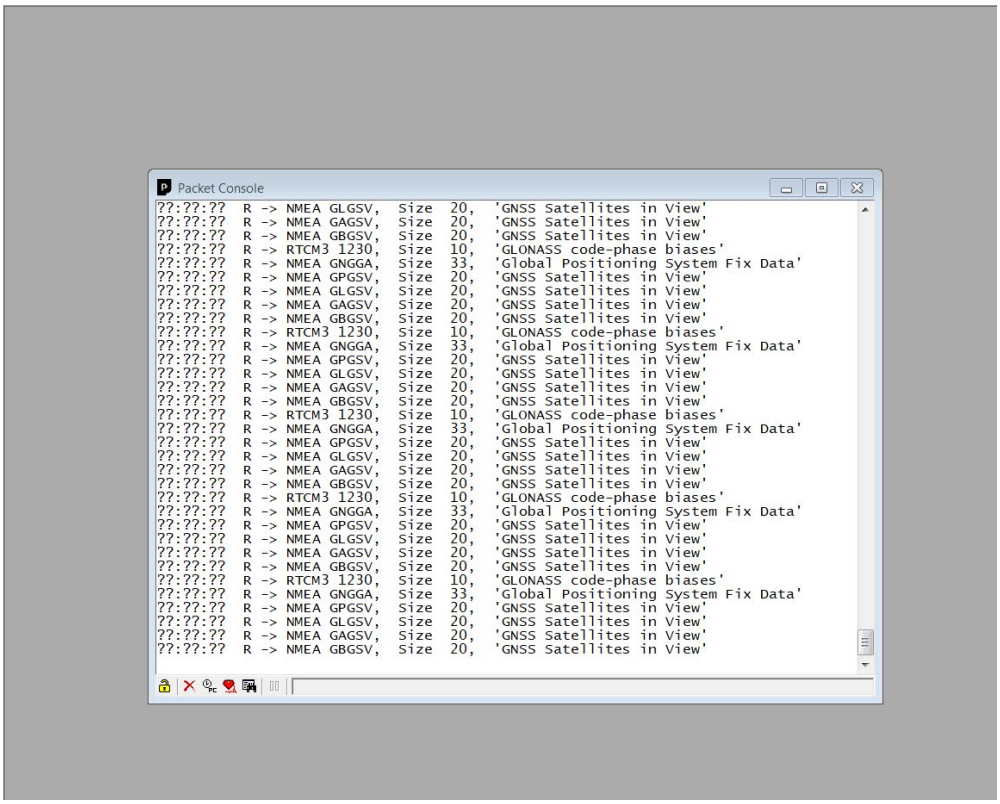
K t n i receiver trong giao di n u-center c• i n.



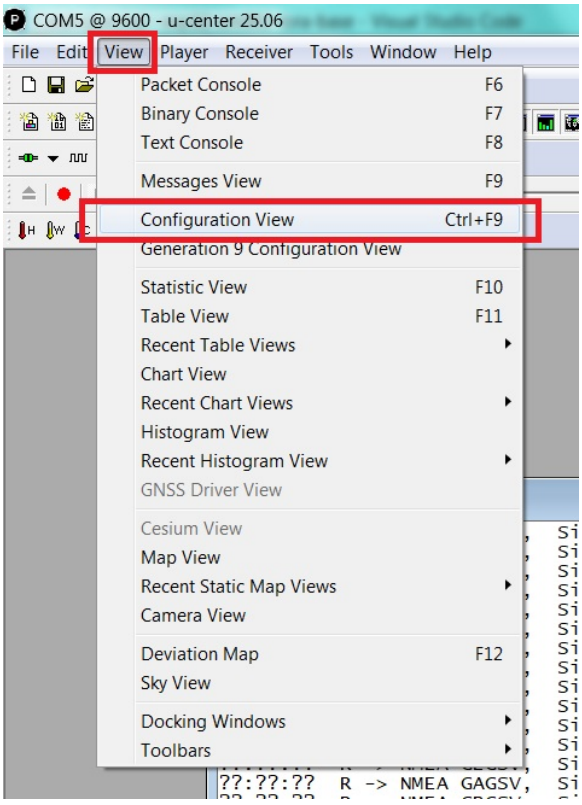
T€ dŽ baud c‹a k t n i receiver.



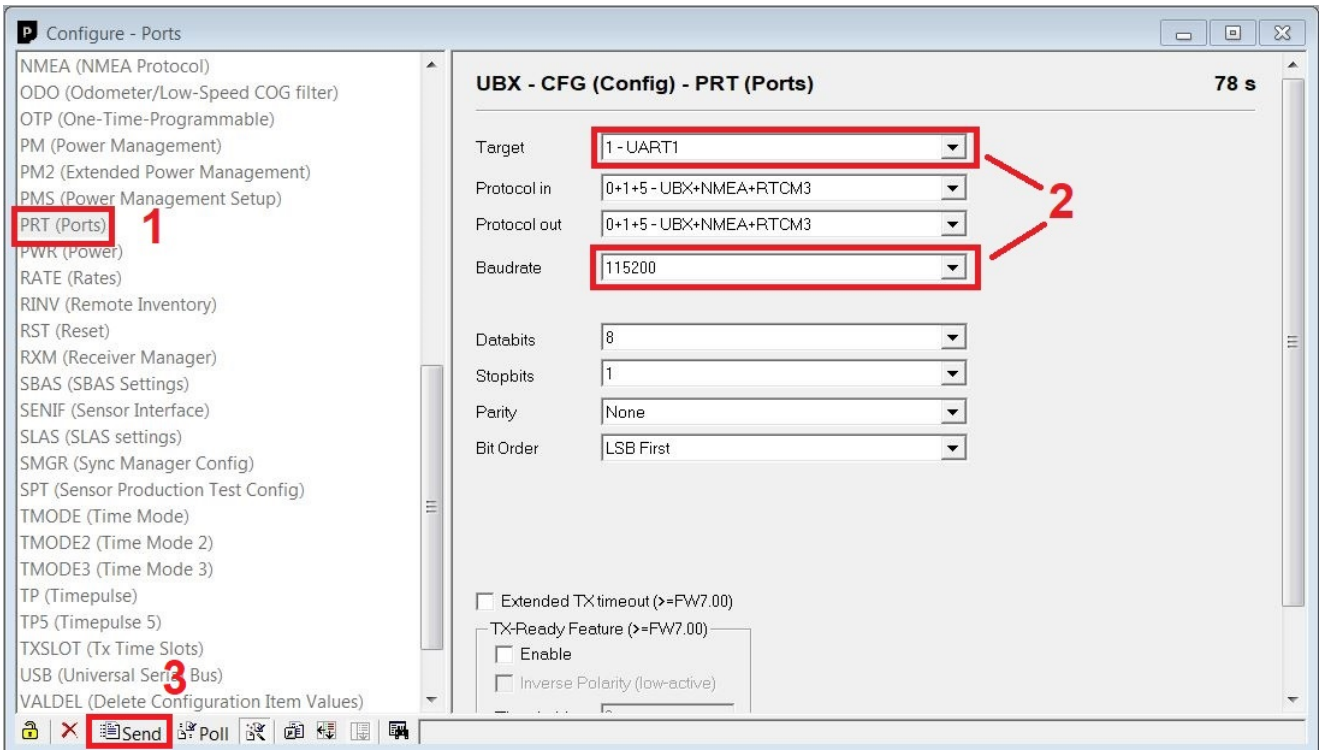
M, c a s• theo d™i b n tin receiver.



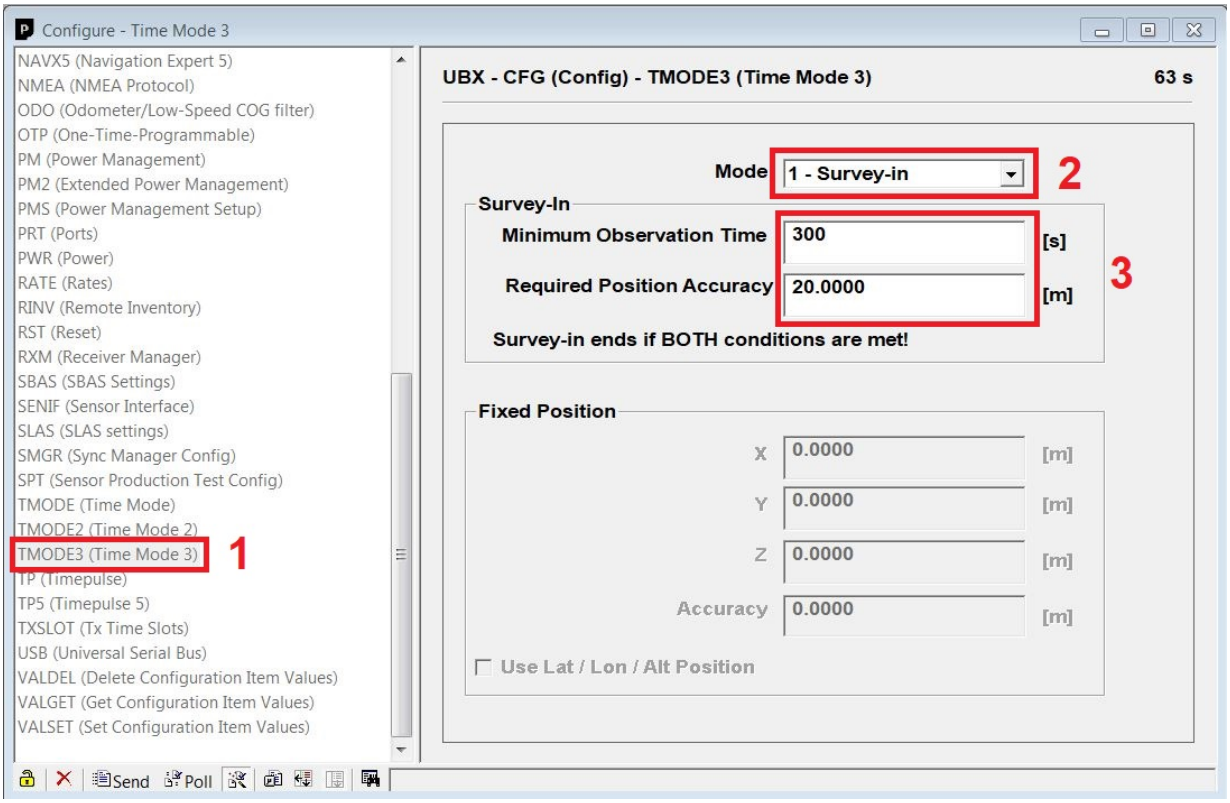
Packet Console hi n th c c lo i g...i nh n ”c.



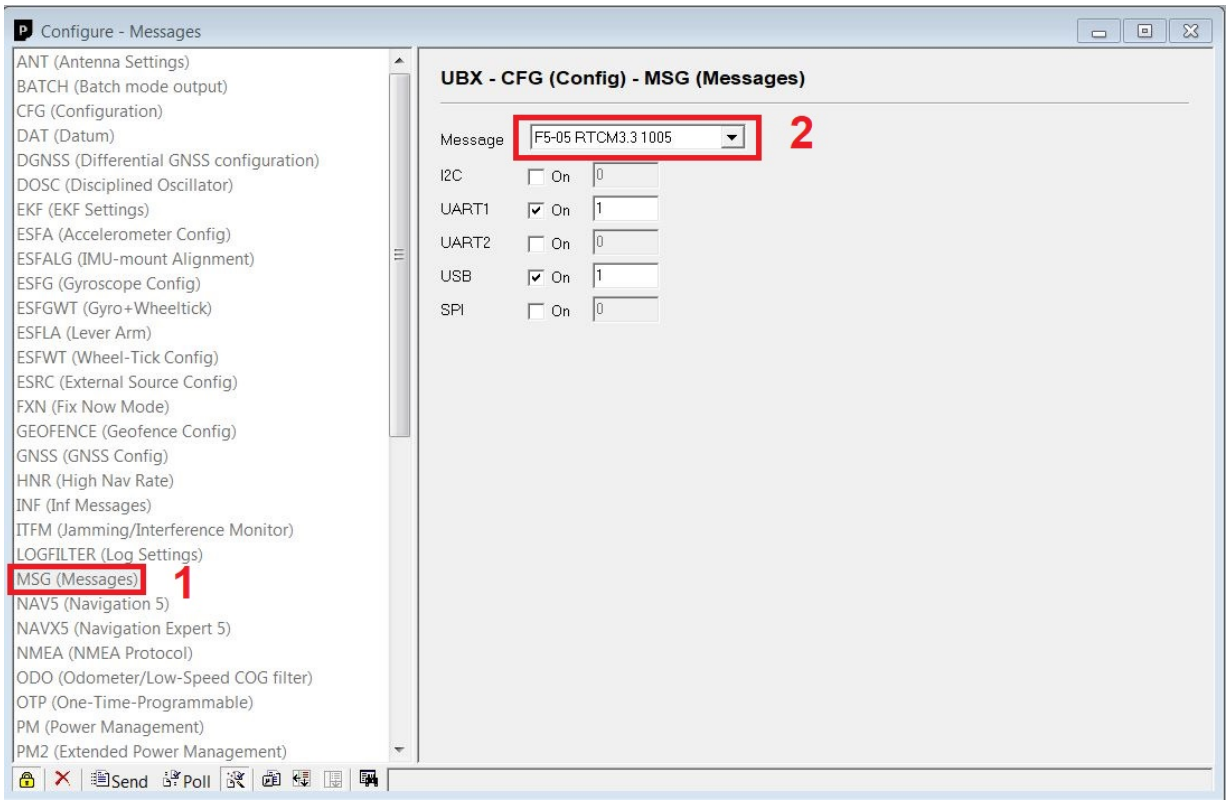
M , giao di n c u h nh receiver.



Ch n c•ng UART v baud c‹a receiver.



C u h nh Time Mode/survey-in.



C u h nh output b n tin tr n t ng c•ng.

9. X l s^ c

D u hi u	Ki„m tra v h ng x l
Kh ng th y log console	ƒ‡ng c•ng USB/driver, 115200 baud, ngu†n v c p truy n d li u.
UNKNOWN/INVALID, kh ng c… GGA	RX/TX u ch›o, chung GND, ‡ng c•ng receiver, ‡ng baud; ki m tra tr€c ti p receiver khi c n.
MQTT ”c nh ng CORS ch a qu n l– ”c	Broker/topic theo serial, thi t b „ nh n/c p v license; ki m tra command/result thay v ch• nh n k t n i TCP.
WebSocket kh ng k t n i	ƒ‡ng host/port endpoint thi t b ; firmware hi n dŠng ws, kh ng ph i wss. MQTT v‰n l k nh i u khi n kh c.
C… NMEA nh ng RTCM b ng 0	ƒ‡ng ch base, survey „ xong, ‡ng output UART v b n tin RTCM „ b t.
C… RTCM UART nh ng caster kh ng nh n	ƒ‡ng source account/mountpoint/version NTRIP; server ”c b t, kh ng kh…a v ‹ license/provision.
C… hai k t n i cŠng t i kho n rover	X c minh client n o o, client n o thu c d ch v monitor/decode tr n caster; kh ng x…a t i kho n khi ch a x c nh.
Auto base kh ng FIX	T i kho n rover, mountpoint u v o c… d li u, GGA m i, c•ng RTCM input, anten/ che khu t v kho ng c ch tr m.
Auto base thi u m‰u	Ki m tra FIX c… duy tr , GGA c… m i v ‹ m‰u h ”p l trong c a s• l y m‰u.
C u h nh GNSS b o l“i	ƒ c ACK/NAK v CFG-TMODE3 readback khi chuy n rover; ki m tra phi n b n receiver h“ tr ” l nh ang dŠng.
Kh , i ng l i v‰n dŠng c u h nh c˜	ƒ i chi u service ang c p t CORS v i c u h nh ATG; ki m tra l u NVS v k t qu tri n khai d ch v .
T a l ch dŠ rover FIX	H quy chi u, cao, chi u cao anten, t a base v ph›p chuy n VN2000; ki m tra tr n m c c l p.

Khi g i y u c u h“ tr ”: cung c p serial, phi n b n, th•i i m l“i, phase/l– do workflow, lo i receiver, baud/c•ng n i, server n o l“i v log „ che m t khŒu/token. Kh ng g i l i reset nhi u l n khi ch a x c nh l“i.

10. Nghi m thu v b o tr

1. Ch p/ghi nh n ‡ng u d‘y, ngu†n, anten, SIM/APN v serial ”c c p.
2. T ’t/b t ngu†n ki m tra thi t b n p ‡ng nh n d ng, d ch v , lo i GNSS v c u h nh base.
3. V i fixed/survey: i chi u t a base v b n tin tham chi u sau khi receiver •n nh.
4. V i auto base: ki m tra t ng b c chuy n rover, nh n c i ch nh, RTK_FIXED, s m‰u, chuy n base v ph c h†i output.
5. DŠng rover c l p k t n i ‡ng mountpoint, ki m tra tu•i c i ch nh, FIX v i m m c „ bi t.
6. Ki m tra ri ng server 1/2, lu n ph t/theo y u c u v kh…a/m , t ng server; kh…a kh ng ”c b fail-open b• qua.
7. Th m t m ng r†i ph c h†i; ki m tra k t qu th€c t , kh ng ch• Ÿn ho c nh„n online.
8. Sau khi di chuy n anten, thay receiver/firmware ho c thay h t a , th€c hi n l i ph n nghi m thu li n quan.

Kh ng c… thao t c o ph n c ng t€ ng n o ”c th€c hi n ch• b , i vi c c ho c xu t b n t i li u n y. B n hi n t i „ i chi u code; nghi m thu ngo i hi n tr •ng do •n v tri n khai ghi nh n ri ng.

T i li u tham chi u

- u-blox ZED-F9P Integration Manual: nguy n t ’c survey-in, fixed v c•ng receiver; c n ch n ‡ng phi n b n t i li u cho thi t b th€c t .